

The Cost of Evidence in Repeated Neural Growth

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Abstract

Can information about what a neural network processes guide repeated structural changes during learning? We investigate a distinction between generating a useful operation and obtaining sufficient evidence to commit it. A trainable shallow network repeatedly proposes an added or split unit, fits both growth and continuation branches, and uses fresh bounded-loss comparisons to grow, continue, or retain its current parameters. A conditional finite-sample argument protects the entire deployed sequence, while allowing it to remain small. The central analysis relates the evidence required for a Brier-risk comparison to prediction movement S and gain Δ , with a worst-case testing cost proportional to S/Δ^2 . An explicit stationary ReLU-module sequence then separates a fresh paired-loss-only audit interface, which requires order K^2/D_0 labels for K powered decisions, from precommitted shared evidence, which saves a factor K up to the displayed logarithmic terms. This is a restricted evidence-handling result, not a lower bound for all neural learning. Ordinary-initialization experiments record eight decisions per trajectory, compare residual, shuffled, random, covariance, and curvature proposals, and account for rejected fits, audit labels, and fixed-capacity alternatives. They establish repeated trained growth in some nonlinear regimes and expose conservative stopping, proposal attribution failures, and the value of giving fixed networks the same label pool. A separately frozen new-seed follow-up replaces the conservative confidence bound with an established betting test, enabling repeated growth at the smaller audit budget while retaining finite-sample validity; random proposals still remain competitive. The supported contribution is a theory and measurement of action evidence, rather than a new universally effective information-growth score.

Contents

1	The question after the earlier phases	2
2	Nearest results and the scope of novelty	3
3	Prediction movement, gain, and the evidence needed to grow	3
4	The cost of resetting evidence across structural decisions	5
5	A trainable process with repeated commitments	8
6	Experiments on ordinary training histories	10
7	Constructive follow-up: improve the gate before blaming the data	16
8	Interpretation and unresolved scientific questions	19
9	Reproducibility and claim status	20
A	Why a rejected singleton need not mean sufficient structure	20

1 The question after the earlier phases

The original curiosity is whether information about what neural units represent or process can guide autonomous structural adaptation. Entropy and splitting motivate possible answers, but neither determines which operation will improve an actual learner. We study a stationary task with a continuing training history. The network receives repeated opportunities to change its width, and its decisions depend on fitted candidates and observed labels. An opportunity schedule is specified externally; growth at that opportunity is not forced. No claim about nonstationary continual learning or semantic specialization is made.

The preserved Phase I project concerns incompatible finite encoders under nested refinement. It neither equates encoder states with neurons nor prices neural optimization. Phase II separates represented information, controller observations, and realizable splits. Its empirical advance is one structural event with an optional heuristic holdout gate. At a designed checkpoint, curvature helps under a short optimization budget; the advantage nearly disappears under longer training. Its operation-aligned quantized conditional information score has no resolved benefit in three reported tasks, while a particular entropy proxy does help the localized task. These mixed findings motivate independent reformulation, not a premise that information can never help.

We ask a narrower operational question with a consequential sequence-level implication: *how does the prediction change caused by a trained action determine the evidence needed to accept it, and what does the learner lose by treating every structural decision as a separate audit?* A useful feature generator and a statistically well-founded commitment rule are different components. Either can limit progress. More information in a candidate does not by itself prove realized benefit after decoder fitting, and a rejected candidate does not establish that the architecture is adequate.

This phase makes four additions to the existing project. First, it implements repeated ordinary-start neural training with growth, continuation, and parameter retention. Second, it supplies complete sequence-valid acceptance arguments under the implemented fresh-data protocol. Third, it gives a quantitative movement/testing analysis and an explicit neural reset-versus-reuse separation, including the escape that limits the lower bound. Fourth, it performs controls that test whether proposal information adds value once training, capacity, labels, and candidate work are visible. Theorem statements and empirical comparisons retain their different scopes throughout.

1.1 Independent exploration and the starting audit

Independent investigations considered residualized features, tangent-space deficiency, nonlinear splitting, observation limitations, audit costs, and complementary action bundles. Residual fitting and tangent projection have close existing formulations. The evidence-cost direction was selected because it makes a testable prediction about an entire learning history and identifies a resource that one-event demonstrations omit. The complementarity direction is retained as a diagnostic boundary in Appendix A. It prevents interpreting conservative no-growth as an optimal architecture.

The workspace contains both original manuscripts, code, and raw results. The supplied PDF hashes match the Phase III handoff exactly. Original-code replays of five representative Phase II regimes reproduce their saved outputs, and independent finite-difference checks confirm the relevant training and function-preserving cloning implementation. No Phase I or II artifact is overwritten. These are load-bearing checks, not a claim that every inherited theorem was formally certified. The new theory does not depend on Phase II's singular high-order split construction.

2 Nearest results and the scope of novelty

Cascade-Correlation already trains candidate units against residual error, adds them progressively, and freezes their incoming weights [5]. AdaNet explicitly compares learned subnetworks using loss plus complexity and stops when its objective does not improve [2, Section 5.2, Figure 3]. Repeated growth and the option to decline it are therefore not new in themselves. Our covariance control is Cascade-inspired; it is not an exact reproduction of Cascade-Correlation’s architecture, frozen weights, or optimizer. All parameters in our grown network remain trainable.

Splitting steepest descent derives the residual-weighted splitting matrix and a minimum-eigenvector operation [11, Section 2.1, Definition 2.1]. We implement that established directional principle for Brier loss, followed by our common finite-radius training and gate. Growing Tiny Networks defines expressivity bottlenecks by projecting desired activity changes onto attainable directions and derives SVD-based neuron initializations [9, Definition 3.1, Propositions 3.1–3.3]. Its Appendix E.5 also explicitly accounts for controller computations. Accordingly, a residual score or a neural growth-cost ledger cannot alone be claimed as our novelty.

Empirical Bernstein concentration and sequential KL change of measure are established tools [7, 8]. Adaptive evaluation has dedicated solutions, including the Ladder leaderboard procedure [1]; repeatedly consulting an ordinary holdout is not validated merely by calling it validation. A recent representation-adequacy study already develops sequential certification in a task-loss currency under fixed representation and observation kernels [6, Section 3, Theorems 8–9]; its Section 4.3 explicitly leaves representation repair outside those guarantees. We do not claim the first information-cost theory of representation certification.

The additional mathematical content here is the explicit movement/gain testing specialization and a stationary neural-module sequence with a proved separation between two evidence interfaces. The basic ingredients, including nested Bayes projection, are credited. The neural-module theorem does not solve general representation repair and does not apply its lower bound to arbitrary supervised training histories. Its value is to show precisely what must be retained or shared to avoid an otherwise repeated evidence cost. External novelty review remains appropriate; the source audit rules out broad originality claims without certifying worldwide priority for the narrower composition.

3 Prediction movement, gain, and the evidence needed to grow

The information used here is evidence about a proposed change in task loss. It is not the entropy of a neuron. Two candidates can have the same small positive gain and very different sampling difficulty, because they move predictions by different amounts. We first make this distinction precise, then exhibit a whole sequence for which discarding evidence between structural decisions imposes a quadratic label cost. The latter cost is a property of a specified audit interface; we also prove how it can be avoided. The decomposition and statistical ingredients are classical. Our additional result is their explicit repeated neural realization and the comparison of fresh and shared evidence, with a deliberately bounded novelty claim.

3.1 The geometry of a paired Brier comparison

Let $Y \in \{0, 1\}$ and $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$. Predictions $p, q : \mathcal{X} \rightarrow [0, 1]$ are fixed before independent audit data. Write

$$D = (Y - p(X))^2 - (Y - q(X))^2, \quad h = q - p, \quad \Delta = \mathbb{E}D, \quad S = \mathbb{E}h^2. \quad (1)$$

Positive Δ means that q improves on p ; S measures prediction movement on population inputs.

Proposition 3.1 (Movement and paired variance). *One has*

$$D = h(2Y - p - q), \quad \Delta = 2\mathbb{E}[h(\eta - p)] - S, \quad (2)$$

$$\text{Var}(D) = 4\mathbb{E}[h^2\eta(1 - \eta)] + \text{Var}\{h(2\eta - p - q)\}. \quad (3)$$

In particular, $|D| \leq 1$ and $\text{Var}(D) \leq 4S$. If $a \leq \eta \leq 1 - a$ almost surely, for $0 < a \leq 1/2$, then $\text{Var}(D) \geq 4a(1 - a)S$. If $p = \mathbb{E}[Y \mid \mathcal{A}]$ and $q = \mathbb{E}[Y \mid \mathcal{B}]$ for sigma fields $\mathcal{A} \subset \mathcal{B}$, then $\Delta = S$.

Proof. Expansion of the two squared losses gives (2). Conditional on X , h, p, q are constants, so $\text{Var}(D \mid X) = 4h^2\eta(1 - \eta)$ and $\mathbb{E}[D \mid X] = h(2\eta - p - q)$. The conditional-variance identity proves (3). Both squared losses lie in $[0, 1]$, giving $|D| \leq 1$; also $|2Y - p - q| \leq 2$, hence $\text{Var}(D) \leq \mathbb{E}D^2 \leq 4S$. The noise lower bound follows from $\eta(1 - \eta) \geq a(1 - a)$. Finally, $q - p$ is $\mathcal{B}$ -measurable and $\mathbb{E}[Y - q \mid \mathcal{B}] = 0$, so $\mathbb{E}[(Y - q)(q - p)] = 0$. Substituting $Y - p = (Y - q) + (q - p)$ in the difference of squared losses gives $\Delta = S$. $\square$

The final assertion is the familiar projection identity for nested Bayes refits. A trained nonlinear branch need not satisfy it. Its gain can be small because two sizable terms in (2) nearly cancel. The ratio S/Δ , when the population gain is positive, records this discrepancy. It is a diagnostic, not a universal control objective.

Proposition 3.2 (A movement-dependent sufficient audit size). *Suppose an upper bound $S \leq \bar{S}$ is fixed independently of the audit. For n iid paired observations and $u > 0$, define*

$$r_n = \sqrt{\frac{8\bar{S}u}{n}} + \frac{4u}{3n}.$$

Each tail probability $\mathbb{P}(\bar{D}_n - \Delta > r_n)$ and $\mathbb{P}(\Delta - \bar{D}_n > r_n)$ is at most e^{-u} . The test $\bar{D}_n > r_n$ therefore has false acceptance probability at most e^{-u} for $\Delta \leq 0$. When $\Delta > 0$, the condition

$$n > \max \left\{ \frac{128\bar{S}u}{\Delta^2}, \frac{16u}{3\Delta} \right\} \quad (4)$$

gives power at least $1 - e^{-u}$.

Proof. Put $Z = D - \Delta$ and $v = \mathbb{E}Z^2$. Then $|Z| \leq 2$. If $v = 0$, D is almost surely constant and both tail bounds hold directly. Assume henceforth $v > 0$. For $0 \leq \lambda < 3/2$, expanding the exponential and using $\mathbb{E}|Z|^k \leq 2^{k-2}v$ and $k! \geq 2 \cdot 3^{k-2}$ for $k \geq 2$ yields

$$\log \mathbb{E}e^{\lambda Z} \leq \frac{\lambda^2 v}{2(1 - 2\lambda/3)}.$$

Apply Markov's inequality to the sum of n independent copies and take $\lambda = r/(v + 2r/3)$ to obtain

$$\mathbb{P}(\bar{Z}_n \geq r) \leq \exp \left\{ -\frac{nr^2}{2(v + 2r/3)} \right\}.$$

For $r = \sqrt{2vu/n} + 4u/(3n)$ the exponent has magnitude at least u . Use $v \leq 4\bar{S}$ and repeat for $-Z$. Under (4), the square-root and linear terms of r_n are each smaller than $\Delta/4$, so $2r_n < \Delta$. Except on the lower-tail failure event, $\bar{D}_n \geq \Delta - r_n > r_n$. $\square$

This is a standard Bernstein argument with an explicit movement bound. Replacing $\bar{S}$ by its same-sample empirical value without a confidence bound would not satisfy the proposition. The empirical gate instead uses an established empirical Bernstein inequality [8]. For nested Bayes refits $S = \Delta$, so the leading sufficient sample order is $1/\Delta$; for unaligned changes it is S/Δ^2 .

3.2 A matching pairwise lower-bound witness

Write $\text{kl}(r, s) = r \log(r/s) + (1 - r) \log\{(1 - r)/(1 - s)\}$. For $\alpha, \beta \in (0, 1)$ with $\alpha + \beta < 1$, let $b_{\alpha, \beta} = \text{kl}(1 - \beta, \alpha)$.

Lemma 3.3 (Sequential testing, established change of measure). *Let P, Q be finite observation laws with positive probabilities. A randomized procedure observes iid draws, stops at a common stopping rule N , and outputs an event A determined by its stopped transcript. There is no initial information whose distribution differs between the laws. If $P(A) \geq 1 - \beta$ and $Q(A) \leq \alpha$, then*

$$\mathbb{E}_P N \text{KL}(P, Q) \geq b_{\alpha, \beta}.$$

Proof. If $\mathbb{E}_P N = \infty$ there is nothing to prove. Otherwise the stopped log-likelihood ratio is $L_N = \sum_{i \geq 1} 1_{\{N \geq i\}} \log\{P(Z_i)/Q(Z_i)\}$. The increments are bounded, and the sum of their absolute expectations is finite because $\mathbb{E}_P N < \infty$. The indicator is determined before draw i , whose log-likelihood increment has mean $\text{KL}(P, Q)$. Thus $\mathbb{E}_P L_N = \mathbb{E}_P N \text{KL}(P, Q)$. The common stopping and randomization rules contribute no extra factor to the likelihood ratio of the stopped transcript. Its KL divergence is therefore $\mathbb{E}_P L_N$. Data processing to 1_A bounds this below by $\text{kl}(P(A), Q(A))$. For $r > s$, binary KL increases with r and decreases with s , as follows by differentiating its displayed formula. Hence this last divergence is at least $b_{\alpha, \beta}$. $\square$

The lemma is the one-observation-channel case of the established change-of-measure result of Kaufmann, Cappé, and Garivier [7, Lemma 1 and Appendix A.1]; its proof is included to make the information restriction inspectable.

Theorem 3.4 (A movement-dependent necessary audit size). *For every $0 < d \leq 1/2$ and $0 < \Delta \leq d/2$, there are two constant predictions with movement $S = d^2$ and two Bernoulli label laws such that the proposed change is risk-neutral under the first law and improves Brier risk by Δ under the second. Any audit with false acceptance at most α under the first and power at least $1 - \beta$ under the second, with no initial distinguishing information, satisfies*

$$\mathbb{E}_1 N \geq b_{\alpha, \beta} \frac{S}{\Delta^2}.$$

Proof. Use a one-point input space and predictions $p = (1 - d)/2$, $q = (1 + d)/2$. Let P_0 have label mean $1/2$ and P_1 mean $1/2 + \Delta/(2d) \leq 3/4$. Then $D = d(2Y - 1)$, so the gains are zero and Δ , while $S = d^2$. The inequality $\log u \leq u - 1$ applied to both Bernoulli KL terms gives $\text{kl}(r, s) \leq (r - s)^2/[s(1 - s)]$. Consequently $\text{KL}(P_1, P_0) \leq \Delta^2/d^2$. Lemma 3.3 proves the claim. $\square$

This is a worst-case pair construction with fixed predictions. It does not assert that every neural pair with the same movement and gain has this difficulty. In particular, the gain alone gives no universal sample bound: some restricted noiseless comparisons have deterministic positive loss differences. Also, a gate that always rejects is safe without any samples; the lower bound depends essentially on the stated power requirement.

4 The cost of resetting evidence across structural decisions

We now give a stationary neural sequence for which the event cost is sharp and all gains add to a fixed risk budget. This is an audit theorem with supplied modules. It is separate from the experiments that must learn their own candidate directions and retrain all parameters.

4.1 Population, neural actions, and observation interface

Fix an even $K \geq 2$, $0 < \varepsilon \leq 1/4$, and known alternating signs $s_t = (-1)^t$. Let X be uniform on $\{1, \dots, K\}$ and

$$\mathbb{P}(Y = 1 \mid X = t) = \frac{1}{2} + s_t \theta_t, \quad 0 \leq \theta_t \leq \varepsilon. \quad (5)$$

With $\rho(z) = \max\{z, 0\}$, define a three-ReLU module

$$\phi_t(x) = \rho(x - t + 1) - 2\rho(x - t) + \rho(x - t - 1). \quad (6)$$

At integer inputs $\phi_t(j) = 1_{\{j=t\}}$. An installed set A predicts $f_A = 1/2 + \varepsilon \sum_{t \in A} s_t \phi_t$, within $[1/4, 3/4]$ on the population support. The initial predictor is $f_\emptyset = 1/2$. Each action adds three actual hidden ReLUs. Their knots and amplitude are supplied; this construction gives no guarantee that SGD finds them.

At opportunity t , module t is proposed, whether or not earlier modules were accepted. The controller may retain the existing network. Its paired loss observation is

$$D_t = 1_{\{X=t\}}(2s_t\varepsilon(Y - 1/2) - \varepsilon^2), \quad g_t = \mathbb{E}D_t = \frac{2\varepsilon\theta_t - \varepsilon^2}{K}. \quad (7)$$

These expressions are independent of earlier acceptance decisions. The known signs are part of the supplied candidate specification. When all $\theta_t = \varepsilon$, the Bayes prediction alternates across inputs, so an intercept alone cannot remove the excess risk. No claim of minimal neural width is made: the theorem fixes the local-module action family.

The *fresh paired-audit interface* supplies a new iid population stream at each opportunity and reveals only the current D_t . The controller may retain all previous paired observations, decisions, and sampling times. It does not retain labels at inputs on which the current action makes no prediction change, nor otherwise acquire data outside this interface. Each audit may stop sequentially using the allowed history and environment-independent randomness. Its sample count N_t includes every labelled population example drawn, including zero paired differences. Assume the following guarantees conditional on every possible earlier allowed transcript:

1. false acceptance is at most α whenever $g_t \leq 0$;
2. acceptance probability is at least $1 - \beta$ if $\theta_t = \varepsilon$;
3. the stopping rule is almost surely finite under the tested laws, with finite expectation under the all-positive law (otherwise the lower bound is immediate).

Here $\alpha, \beta \in (0, 1)$ and $\alpha + \beta < 1$. These are distribution-uniform audit requirements, not assertions about the power of the implemented neural gate.

Theorem 4.1 (Serial evidence cost and a matching audit). *Under (5), take $\theta_t = \varepsilon$ for every t . Every candidate has gain and movement $\Delta_t = S_t = \varepsilon^2/K$. The initial excess Brier risk above the Bayes predictor is $D_0 = \varepsilon^2$. Every fresh paired-audit controller satisfying the preceding requirements obeys*

$$\mathbb{E} \sum_{t=1}^K N_t \geq \frac{K^2}{2\varepsilon^2} b_{\alpha, \beta} = \frac{K^2}{2D_0} b_{\alpha, \beta}. \quad (8)$$

The expected number accepted is at least $K(1 - \beta)$, and expected risk reduction is at least $(1 - \beta)D_0$. Conversely, the same interface admits an audit satisfying these guarantees with

$$\mathbb{E} \sum_{t=1}^K N_t = K^2 \left\lceil \frac{8}{\varepsilon^2} \log \frac{1}{\min\{\alpha, \beta\}} \right\rceil. \quad (9)$$

Proof. The identity $\mathbb{E}(Y - f)^2 = \mathbb{E}[\eta(1 - \eta)] + \mathbb{E}(\eta - f)^2$ shows that each module removes exactly ε^2/K excess risk. All K remove $D_0 = \varepsilon^2$; prediction movement is also ε^2/K .

Fix t and compare the all-positive environment P to Q_t , which only replaces θ_t by $\varepsilon/2$. The stage- t move is risk-neutral under Q_t . Every earlier observation D_s , $s < t$, has exactly the same law under both environments: it is zero at $X = t$, while its only label-dependent value occurs at $X = s$, whose label law did not change. Inducting through the common stopping and decision rules shows that the entire earlier allowed transcript has the same law. Conditional on almost every transcript under this common law, no initial information distinguishes P from Q_t .

The signed label $\tilde{Y}_t = \{1 + s_t(2Y - 1)\}/2$, conditional on $X = t$, has mean $1/2 + \theta_t$. Relabelling by this invertible map preserves KL. The one-sample full-pair divergence therefore satisfies

$$\begin{aligned} \text{KL}(P, Q_t) &= \frac{1}{K} \text{kl}(1/2 + \varepsilon, 1/2 + \varepsilon/2) \\ &\leq \frac{1}{K} \frac{(\varepsilon/2)^2}{(1/2 + \varepsilon/2)(1/2 - \varepsilon/2)} \leq \frac{2\varepsilon^2}{K}. \end{aligned}$$

The last denominator is at least $15/64$; the displayed factor two is conservative. Mapping full pairs to D_t cannot increase KL. Apply Lemma 3.3 conditionally on the earlier transcript to obtain $\mathbb{E}_P[N_t \mid \text{history}] \geq Kb_{\alpha,\beta}/(2\varepsilon^2)$. Integration and summation prove (8). Conditional power and linearity of expectation give the remaining lower-bound assertions. The decisions need not be independent.

For the upper bound, let $m = \lceil 8\varepsilon^{-2} \log(1/\min\{\alpha, \beta\}) \rceil$. At stage t , wait for m nonzero paired observations; these are exactly the examples with $X = t$, and either nonzero value identifies Y . Accept if their mean signed label $\tilde{Y}_t$ exceeds $1/2 + 3\varepsilon/4$. For $\theta_t \leq \varepsilon/2$ and for $\theta_t = \varepsilon$, the mean is at least $\varepsilon/4$ from the threshold in the appropriate direction. The Bernoulli Hoeffding bound gives either error at most $e^{-m\varepsilon^2/8} \leq \min\{\alpha, \beta\}$. For completeness, Hoeffding's bound follows by bounding the centered Bernoulli moment generating function by $e^{\lambda^2/8}$, applying Markov's inequality to m independent samples, and optimizing at $\lambda = 4a$ for a deviation a ; the result is e^{-2ma^2} . The waiting time to m occurrences of an event of probability $1/K$ has mean Km , being the sum of m geometric waiting times of mean K . Summing over opportunities proves (9). $\square$

For fixed small error probabilities, the cost is quadratic in the number of useful proposed modules at fixed total removable risk. If all K decisions must be simultaneously correct with probability at least $1 - \delta$, taking $\alpha = \beta = \delta/K$ supplies that guarantee and adds the usual logarithmic factor. The lower bound is a change-of-measure argument over an entire stationary neural sequence, not a repeated union bound. Its limitation is equally central: the observation interface discards off-action information, and the modules are prescribed.

4.2 A constructive escape through shared evidence

Theorem 4.2 (Precommitted reuse removes a factor K). *In the same population, fix all modules before seeing labels and assume each θ_t is either at most $\varepsilon/2$ or exactly ε . One common batch of*

$$n = \left\lceil \frac{16K}{\varepsilon^2} \log \frac{2K}{\delta} \right\rceil \quad (10)$$

iid labelled pairs is sufficient to accept every positive separated alternative and reject every nonpositive proposed change with probability at least $1 - \delta$. The accepted modules can be installed successively. On the subfamily $\theta_t \in \{0, \varepsilon\}$ the resulting network contains exactly the needed modules on this event, including no growth when all $\theta_t = 0$.

Proof. For each coordinate, use its signed labels $\tilde{Y}_t$ in the common batch and accept if their mean exceeds $1/2 + 3\varepsilon/4$. Let C_t count examples at coordinate t and put $m_0 = n/(2K)$. A multiplicative binomial bound gives $\mathbb{P}(C_t < m_0) \leq e^{-n/(8K)}$. It follows directly from Markov's

inequality applied to $e^{-\lambda C_t}$: the binomial generating function is at most $\exp\{(n/K)(e^{-\lambda} - 1)\}$; taking $\lambda = \log 2$ gives an exponent no greater than $-n/(8K)$ at $C_t = n/(2K)$. Conditional on counts and $C_t \geq m_0$, Hoeffding’s bound makes the test error at most $e^{-m_0 \varepsilon^2/8} \leq \delta/(2K)$. Also $e^{-n/(8K)} \leq \delta/(2K)$ by (10) and $\varepsilon \leq 1/4$. A union bound over the K count failures and K test errors proves the claim. Locality makes each comparison independent of which other modules were installed, so sequential installation does not change the statements being tested. $\square$

The reuse theorem is valid because the local candidates and tests were fixed independently of the labels. It does not validate arbitrary holdout reuse after neural fitting or feedback-dependent proposal changes; adaptive evaluation is a separate problem with established solutions and limitations [1]. Alternatively, screening inputs without requesting labels and labelling only $X = t$ reduces the separate tests to Km total labels. It still takes K^2m expected input inspections unless direct conditional input sampling is available. These resources should not be conflated.

The exact sequence also explains why a rejected operation is not an adequacy certificate. At $\theta_t = \varepsilon/2$, adding amplitude ε is risk-neutral, while adding amplitude $\varepsilon/2$ would improve risk. Our no-growth statement concerns a specified action family; it certifies complete adequacy only on the subfamily $\theta_t \in \{0, \varepsilon\}$ stated in Theorem 4.2.

4.3 What transfers to a learned controller

The elementary implication

$$\sum_{t=1}^K \Delta_t \leq D_0, \quad \mathbb{E} N_t \geq c/\Delta_t \quad \implies \quad \sum_{t=1}^K \mathbb{E} N_t \geq cK^2/D_0$$

follows from Cauchy–Schwarz. The per-event premise is not automatic; Theorem 4.1 supplies an exact realization. For more general comparisons, the relevant terms may be S_t/Δ_t^2 , and their observed values should be reported instead of asserting a universal exponent.

There is an additional information-accounting caveat. If initial history H has different laws under the competing environments, the correct testing constraint is

$$\text{KL}(P_H, Q_H) + \mathbb{E}_P N \text{KL}(P, Q) \geq \text{kl}(P(A), Q(A)), \quad (11)$$

when the new iid stream is independent of H under each environment. This follows from the KL chain rule and the proof of Lemma 3.3; history-dependent sampling replaces the second term by a sum of conditional divergences. Initial training or retained audit data can pay for the decision. Dropping the first term while allowing arbitrary historical information would make a fresh-audit lower bound false. The same labels cannot then be counted repeatedly as new purchases.

Consequently the neural experiments below test movement, realized gain, and audit cost along ordinary trained trajectories; they are not numerical proofs that those trajectories satisfy the local-module lower bound. The observation distinction survives this limitation: failure to accept can mean either an ineffective operation or inadequate evidence for a useful one. Recording both terms is necessary to tell these explanations apart.

5 A trainable process with repeated commitments

5.1 Network, observations, and actions

The implemented classifier is

$$p_\theta(x) = \text{sigmoid} \left(c + \sum_{j=1}^k a_j \tanh(w_j^\top x + b_j) \right), \quad \ell_\theta(x, y) = (y - p_\theta(x))^2. \quad (12)$$

All incoming weights, biases, outgoing weights, and the intercept are trainable. For input dimension d there are $k(d + 2) + 1$ parameters. The sigmoid bounds probabilities and makes the loss lie in $[0, 1]$ without an artificial boundedness assumption or evaluation clipping. We use Brier loss to keep the actual training objective, audit, and movement theory aligned.

The controller is centralized. It sees the fitting inputs and labels, current predictions, residuals, and parameters. It also sees a fresh audit block after both competing branches have been fitted. It receives no teacher probabilities, latent task directions, or evaluator outputs. A learned proposal adds one unit with trained incoming parameters and outgoing weight zero. This preserves the current predictor exactly. The first gradient step can change the new outgoing weight; subsequent steps can train its incoming weights. Splitting instead duplicates a selected unit, halves its outgoing weight, and perturbs the two incoming weight vectors oppositely. All parameters subsequently train without tying. Neither action imposes nested partitions or permanent neuron specialization.

5.2 Proposal generation as a controlled mechanism

For a fitting sample of size n_F , freeze current predictions p_i and residuals $r_i = y_i - p_i$. For a candidate $h_{w,b}(x) = \tanh(w^\top x + b)$, define the probability tangent $z_i = p_i(1 - p_i)h_{w,b}(x_i)$. The primary score is

$$J(w, b) = \frac{(n_F^{-1} \sum_i r_i z_i)^2}{n_F^{-1} \sum_i z_i^2 + 10^{-3}}. \quad (13)$$

It is a regularized one-direction least-squares proxy for a small logit addition, not an exact post-training risk improvement and not mutual information. Its denominator and the sigmoid derivative matter; the score has no asserted scale or reparameterization invariance. Four independent candidate units undergo 60 Adam ascent steps. The largest training score selects one unit, and only that grown network receives the subsequent full branch fit. We charge the optimization of all four candidate units.

The shuffled control permutes the frozen residual vector once per opportunity and uses the same score, candidate initializations, ascent steps, and branch training. It isolates meaningful input-residual pairing from that proposal search workload. Random addition chooses an untrained candidate from the same initial pool; it uses the same gate and full branch training but costs less proposal optimization. The covariance control maximizes $[n_F^{-1} \sum_i r_i (h_i - \bar{h})]^2$; it uses the same four-candidate and 60-step machinery. Squaring the covariance has the same maximizers as its absolute value, but our shallow architecture and all-parameter retraining differ from full Cascade-Correlation.

The SSD control selects the lowest eigenvalue across matrices

$$\hat{S}_j = \frac{1}{n_F} \sum_i 2(p_i - y_i)p_i(1 - p_i)a_j \tanh''(w_j^\top x_i + b_j)x_i x_i^\top. \quad (14)$$

The associated incoming direction gives a radius-0.5 symmetric split. Biases are not perturbed initially. The empirical gate, rather than the sign of this infinitesimal matrix alone, decides whether the finitely trained split is retained. This is a controlled use of SSD, not a claim to reproduce every choice in its original training procedure.

5.3 Growth, continuation, or holding the checkpoint

At opportunity t , let I_t be the incumbent. Independently of the upcoming audit, fit a same-width continuation C_t and one proposed growth G_t from the same incumbent, each for 100 Adam steps on the persistent fitting data. Optimizer moments reset to zero for both branches. After a decision, only the selected parameters become the next incumbent. Holding retains them exactly, so an unsuccessful branch cannot silently alter the deployed model.

On n_A fresh iid audit examples compute improvements for the three ordered pairs (I_t, C_t) , (I_t, G_t) , and (C_t, G_t) . For paired differences D_i and sample variance s_D^2 with denominator $n_A - 1$, define

$$L(D) = \bar{D} - \sqrt{\frac{2s_D^2 \log(2/\alpha_t)}{n_A}} - \frac{14 \log(2/\alpha_t)}{3(n_A - 1)}. \quad (15)$$

For eight opportunities and total error $\delta = 0.05$, use $\alpha_t = \delta/(3 \cdot 8)$. With $\tau = 0.0005$, choose

$$I_{t+1} = \begin{cases} G_t, & L(I_t - G_t) > \tau \text{ and } L(C_t - G_t) > \tau, \\ C_t, & \text{otherwise, if } L(I_t - C_t) > 0, \\ I_t, & \text{otherwise.} \end{cases} \quad (16)$$

Here $L(A - B)$ means the lower bound applied to loss of A minus loss of B . The positive growth margin expresses a minimum improvement requirement; it is not a fitted biological threshold or a monetary price.

Proposition 5.1 (Conditional validity over the deployed history). *At each opportunity, suppose all proposal generation, branch fitting, and choice of audit size occur before an independent iid audit block is revealed, with each audit size at least two. Losses lie in $[0, 1]$. Allocate comparison error probabilities predictably with total at most δ , over a finite or countable sequence. Then, with probability at least $1 - \delta$, every accepted continuation in (16) has lower population risk than its incumbent, every accepted growth improves on both incumbent and fitted continuation by more than τ , and every hold leaves population risk unchanged. Therefore the deployed incumbent risk never increases and accepted growth improvements sum to at most the initial risk. No power or architectural adequacy guarantee is implied by a hold.*

Proof. Condition on the complete earlier history, fitting sample, and current proposal randomness. The three predictors are then fixed, and the new audit remains iid. For each pair $D \in [-1, 1]$, apply Theorem 4 of Maurer and Pontil [8] to $(1 - D)/2 \in [0, 1]$. Scaling the sample variance and mean gives exactly $\mathbb{P}(L(D) > \mathbb{E}D \mid \text{history}) \leq \alpha_t$ in (15). The three pairs may use the same block; a union bound does not require independence among them. Conditional failure bounds, integration over histories, and the allocated total give simultaneous coverage at all opportunities. For a countable sequence use the increasing union of finite failure events. On the covered event the inequalities in (16) imply the stated risk improvements; holding preserves the predictor. Summing nonnegative risk decrements proves the budget claim. $\square$

This guarantee is per specified controller trajectory. It is not a simultaneous guarantee across every experimental seed and method without further allocation. It protects the accepted checkpoints, not the trial branches, warmup, or the later ungated training diagnostic. It compares growth with a continuation from the same incumbent, not with an independently progressing fixed-width learner. The general fresh-data guarantee applies to the actual sigmoid network; the stronger powered-audit lower bound in Theorem 4.1 does not. Our controller has historical fitting labels and is not constrained to the theorem’s paired-loss-only interface.

6 Experiments on ordinary training histories

6.1 Tasks, samples, and the frozen confirmation protocol

Every main task has $X \sim N(0, I_6)$ and sampled binary labels with $\eta(x) = 0.05 + 0.90 \text{ sigmoid}(g(x))$. The four functions are

$$\begin{aligned} g_{\text{null}}(x) &= 0, & g_{\text{linear}}(x) &= 2x_1, \\ g_{\text{additive}}(x) &= 2.5 \tanh(2x_1 + 0.6) - 2 \tanh(2x_2 - 0.5) \\ &\quad + 1.7 \tanh(2x_3 + 0.3) - 1.5 \tanh(2x_4), \\ g_{\text{interaction}}(x) &= 4 \tanh(2x_1) \tanh(2x_2) + 2 \tanh(2x_3) \tanh(2x_4). \end{aligned}$$

These are fixed synthetic mechanism tests with nuisance coordinates, not a broad benchmark suite. Main fitting uses 2,048 iid examples. Every network starts at width two with incoming weights $N(0, 0.7^2)$, biases $N(0, 0.15^2)$, outgoing weights $N(0, 0.35^2)$, and intercept zero, followed by 200 ordinary full-batch Adam updates. All-parameter fitting uses learning rate 0.035; candidate ascent uses 0.04. Adam coefficients are (0.9, 0.999) with denominator constant 10^{-8} . The same rates and steps are applied across tasks; no designed stationary checkpoint is supplied.

Exploration used separate seeds 51000–51002 and 52000–52002. After two pilots and code review, the method, comparison set, outcomes, and 16 new seeds 61000–61015 were recorded before confirmation. The two main finite-bound protocols use 4,096 and 16,384 fresh labels per opportunity at the same fitting sample size. They share fitting data, evaluation inputs, starting parameters, and event-indexed candidate initializations. Their audit blocks are separate draws, not nested prefixes. Within a method, all eight audit blocks are disjoint. No block is returned to its fitting loop. After selection, audit outcome information enters the next incumbent, which is why fresh data are needed for the next conditional argument.

An acceptance ablation uses the empirical means in (16) instead of the lower bounds, with the same τ , data, and residual/shuffled/random proposal rules. It is explicitly heuristic and has no sequence risk guarantee. It does not replace the primary method based on favorable final evaluation. Both protocols and all unfavorable trajectories remain in the raw files.

Evaluation averages $(p_\theta(X) - \eta(X))^2 + \eta(X)(1 - \eta(X))$ over 16,384 independent inputs. It exactly integrates conditional Bernoulli label noise but is Monte Carlo over X , not exact population evaluation. These probabilities are accessed only by evaluator functions after the action is fixed. Independent runs are the replication unit. Reported paired intervals are two-sided 95% Student t intervals over 16 seeds, descriptive and not multiplicity-adjusted. Identical intervals containing zero do not establish equivalence.

6.2 Controls and accounting

We distinguish fitting, audit, candidate optimization, final capacity, and post-training. The retained model has $8k + 1$ parameters; the controller holds an incumbent and fits two branches. A decision opportunity costs both 100-step branch fits even if growth is rejected, plus the candidate pool’s optimization. We record parameter–example updates as an operation proxy: training steps times sample size times parameter count. Candidate incoming optimization contributes $4 \cdot 60 \cdot 7n_F$ at an addition opportunity. SSD matrix formation and eigendecomposition have a separate arithmetic proxy $O(n_Fkd^2 + kd^3)$. It is not silently combined with parameter updates.

Decision-required audit prediction work is reported separately for the three fitted predictors. Additional forwards record prediction movement for scientific diagnosis; elapsed controller times include that instrumentation. They exclude final evaluation and fixed-model controls. Simultaneously running local processes and differing linear algebra costs prevent interpreting these times as matched-hardware efficiency. Peak process memory is not measured; final predictor parameter count is not a claim about controller memory.

An ungated width-two learner and an ordinary fixed width-ten model receive the nominal warmup-plus-one-branch-per-opportunity budget of 1,000 full-batch updates. Holds discard the trial training, so this is not the number of updates embodied in every retained path. Each also receives a separate 600-step continuation. A retrospective fixed-final-capacity model starts from scratch at the width reached by each adaptive run and receives the adaptive controller’s branch-and-probe update proxy. This measures alternative optimization at matched final capacity; knowing that width in advance is diagnostic information, so this comparator is not a deployable automatic-width method. Both it and the grown network are then trained for 600 additional steps without gates to test whether any advantage survives longer optimization. Those tails are outside Proposition 5.1.

The low- and high-audit controllers have access to 34,816 and 133,120 labelled examples in total, despite fitting on only 2,048. A second fixed-width control is therefore granted the

Task	Proposal	$n_A = 4096$	$n_A = 16384$	Width	Grow ≥ 2
Null	Residual	0.2532	0.2532	2.00	0.00
	Shuffled	0.2532	0.2532	2.00	0.00
	Random	0.2532	0.2532	2.00	0.00
	Covariance	0.2532	0.2532	2.00	0.00
	Ssd	0.2532	0.2532	2.00	0.00
Linear	Residual	0.1720	0.1720	2.00	0.00
	Shuffled	0.1720	0.1720	2.00	0.00
	Random	0.1720	0.1720	2.00	0.00
	Covariance	0.1720	0.1720	2.00	0.00
	Ssd	0.1720	0.1720	2.00	0.00
Additive	Residual	0.1472	0.1395	3.25	0.56
	Shuffled	0.1472	0.1427	2.62	0.19
	Random	0.1472	0.1377	3.50	0.62
	Covariance	0.1472	0.1409	3.00	0.44
	Ssd	0.1472	0.1381	3.38	0.62
Interaction	Residual	0.1754	0.1507	5.62	1.00
	Shuffled	0.1749	0.1531	5.25	1.00
	Random	0.1739	0.1475	5.88	1.00
	Covariance	0.1759	0.1498	5.69	1.00
	Ssd	0.2113	0.2071	2.69	0.00

Table 1: Mean final Brier risk over 16 new seeds per task. Width and fraction with at least two accepted growth events refer to the 16,384-label audit. Both columns fit on the same 2,048 examples and use eight fresh audit blocks. Uncertainty for the primary paired comparisons is in Table 2; complete marginal intervals are in the CSV.

entire fitting-plus-audit label pool. It cycles random minibatches of size 256 through that pool using persistent Adam, learning rate 0.015, at the normalized-residual controller’s training/probe parameter–example budget. Both width two and width ten are tested, with actual unique labels and updates logged. This is an end-of-lifetime offline resource-allocation comparison: it receives the full pool at the outset and uses a different competent optimizer schedule. It shows what that label budget can buy; it does not isolate a data-only causal effect. The first pilot’s weaker sampled-pool/reset control is retained and not passed off as this stronger comparator.

6.3 Measured outcomes and attribution

The main confirmation contains 192 task–seed–protocol runs, representing 832 controller trajectories and 6,656 structural opportunities. All completed histories pass independent reconstruction of actions, parameters, resource counts, and terminal risks. Figures 1 and 2 display the full opportunity sequence, including holds. Table 1 reports every main proposal rule, rather than selecting the best-performing rule per task.

Repeated growth and genuine no-growth. At the larger audit budget, the residual method accepts 20 additions across 16 additive-task runs and 58 across 16 interaction-task runs. Nine of 16 additive runs and all 16 interaction runs accept at least two additions. Mean final widths are 3.25 and 5.625; final risks are 0.13948 ± 0.00280 and 0.15067 ± 0.00291 , where uncertainty denotes a 95% interval half-width. Growth and holding occur within the same histories; these are not restarts at independently constructed checkpoints. On both null and linear tasks every primary finite-bound trajectory retains width two and its warmup parameters. This prevents further overfitting but does not recover Bayes risk automatically: null risk is 0.25325, above the known 0.25 Bayes risk because warmup itself was ungated.

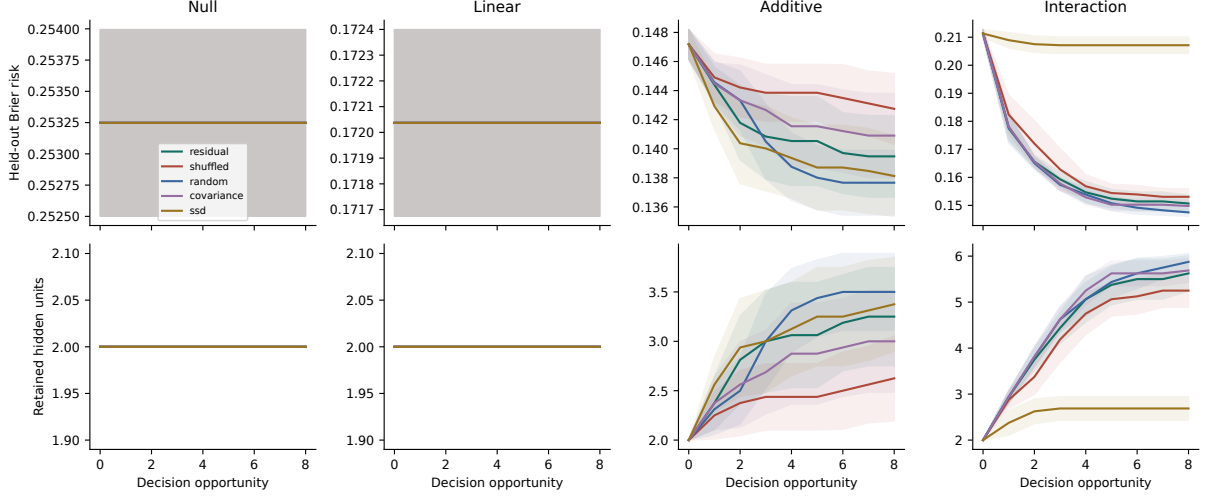


Figure 1: Measured ordinary-start trajectories at 16,384 audit labels per opportunity. Curves are mean retained risk and width; shaded bands are pointwise 95% Student- t intervals across 16 independent seeds. Risk integrates conditional label noise over 16,384 independent evaluation inputs. All methods coincide on null and linear tasks. The same predictor persists through a hold, while trial branches still consume computation.

At 4,096 audit labels, no residual additive run grows and each interaction run grows exactly once, to width three. The interaction risk is 0.17538, compared with 0.15067 at the larger audit budget. Fitting data and proposal rules are fixed across this comparison, so the difference concerns the evidence protocol and its induced history, not extra fitting samples. It nevertheless costs four times as many audit labels. It is not a free improvement in training efficiency.

Proposal information is not a general advantage. Residual minus shuffled risk on the larger-audit additive task is -0.003268 ± 0.002547 . This is a favorable comparison at equal candidate optimization workload, with the caveat of descriptive multiple comparisons. It does not survive the stronger interpretation of beating random proposal addition: residual minus random is $+0.001813 \pm 0.002433$ on additive data and $+0.003168 \pm 0.002128$ on interaction data. Random is significantly better in the latter descriptive paired comparison while spending less on candidate optimization. At the smaller audit budget, interaction residual minus random is also adverse, $+0.001528 \pm 0.000854$. The covariance rule is not reliably separated from the residual rule. SSD is much worse on this interaction task, but its failure cannot establish superiority over all growth principles: it perturbs a different restricted action family and nearly fails to grow. These observations favor retaining proposal alternatives instead of calling the selected residual proxy a validated information principle.

Capacity and optimization controls limit the attribution. Compared with ungated width-two training, residual growth improves mean larger-audit risk by 0.00655 on additive data and 0.05952 on interaction data. Thus the implemented repeated process does realize useful new capacity on these tasks. At retrospectively matched final capacity and training/probe work, however, paired differences are -0.000794 ± 0.001643 and -0.000496 ± 0.002836 : no resolved final-checkpoint advantage. After the 600-step tail the interaction comparison is modestly favorable, -0.002268 ± 0.002154 , while the additive interval spans zero. This isolated tail effect does not establish a general efficiency result. Full-batch training on 2,048 noisy labels can overfit both larger fixed networks and grown ones. The deployed checkpoint and ungated tail therefore must not be conflated.

At a matched training/probe update proxy, an offline width-ten learner given the full

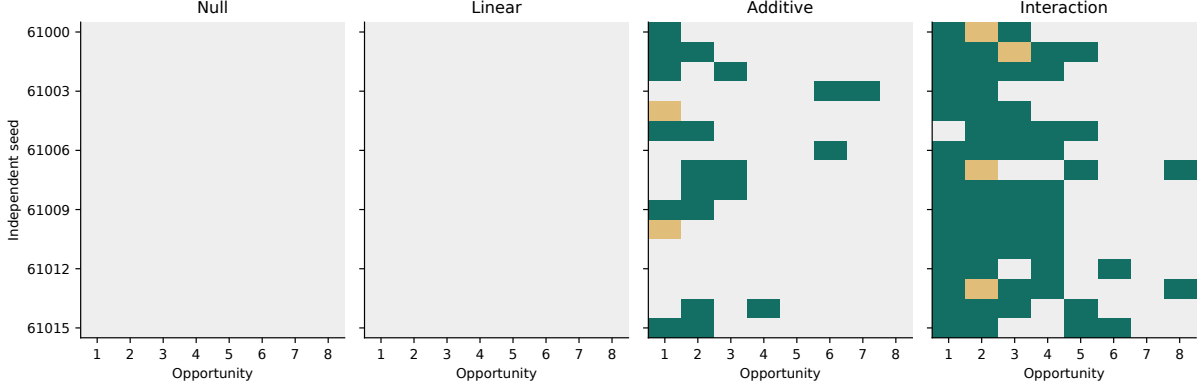


Figure 2: Every primary residual trajectory at the larger audit budget. Green means accepted growth; amber means accepted continuation; light gray means the incumbent parameters are held. Rows are the 16 prespecified seeds, columns the eight decision opportunities. An external opportunity schedule exists, but actions and accepted final capacity are data-dependent.

Comparator	Null	Linear	Additive	Interaction
shuffled	$+0.0000 \pm 0.0000$	$+0.0000 \pm 0.0000$	-0.0033 ± 0.0025	-0.0024 ± 0.0042
random	$+0.0000 \pm 0.0000$	$+0.0000 \pm 0.0000$	$+0.0018 \pm 0.0024$	$+0.0032 \pm 0.0021$
covariance	$+0.0000 \pm 0.0000$	$+0.0000 \pm 0.0000$	-0.0014 ± 0.0019	$+0.0009 \pm 0.0023$
ssd	$+0.0000 \pm 0.0000$	$+0.0000 \pm 0.0000$	$+0.0014 \pm 0.0032$	-0.0565 ± 0.0042
fixed matched	-0.0022 ± 0.0009	-0.0007 ± 0.0004	-0.0008 ± 0.0016	-0.0005 ± 0.0028
fixed matched tail	-0.0007 ± 0.0008	-0.0004 ± 0.0004	-0.0008 ± 0.0010	-0.0023 ± 0.0022
small	-0.0016 ± 0.0006	-0.0001 ± 0.0003	-0.0066 ± 0.0034	-0.0595 ± 0.0031
large	-0.0229 ± 0.0013	-0.0176 ± 0.0012	-0.0120 ± 0.0027	-0.0067 ± 0.0031
large pool	$+0.0030 \pm 0.0008$	$+0.0014 \pm 0.0004$	$+0.0080 \pm 0.0029$	$+0.0104 \pm 0.0025$

Table 2: Residual minus comparator Brier risk at the larger audit budget: paired mean $\pm$ 95% Student-*t* half-width over 16 runs. Negative favors residual growth. “Fixed matched tail” compares both predictors after 600 additional updates; other rows use final retained checkpoints. “Large pool” is width ten trained offline from the full label pool at matched training/probe update proxy, with different minibatch optimization. Intervals are descriptive, not multiplicity-adjusted discoveries.

fitting-plus-audit label pool beats the residual controller on all four tasks. On additive and interaction data, residual minus this comparator is $+0.007953 \pm 0.002875$ and $+0.010387 \pm 0.002514$ respectively. The corresponding fixed-pool risks are 0.13152 and 0.14028. These models use a different minibatch schedule and receive future audit labels at the outset; the result is an alternative end-of-lifetime resource allocation, not a causal estimate of the effect of labels alone. It prevents claiming a predictive-risk advantage at this total label and training budget. The fixed comparator deploys ten units, however, whereas the adaptive models average 3.25 and 5.625; it does not dominate them simultaneously in inference capacity.

An additional, explicitly post-hoc diagnostic addresses this capacity mismatch. For the 64 larger-audit residual runs, a fixed learner uses the same final width and initialization as the existing matched-width comparator, but trains from the whole label pool with the already specified minibatch schedule and adaptive training/probe budget. It actually consumes all 133,120 distinct labels in every run. Residual minus this same-width pool comparator is $+0.003111 \pm 0.000767$ on null data, $+0.001266 \pm 0.000415$ on linear data, $+0.003491 \pm 0.000860$ on additive data, and $+0.002246 \pm 0.001887$ on interaction data. This removes the deployed-width mismatch and reinforces the resource-allocation limitation. The comparator still receives labels offline, uses a different optimizer schedule, and knows final width retrospectively. It is a diagnostic added after review, not a preregistered primary finding or an online dominance theorem.

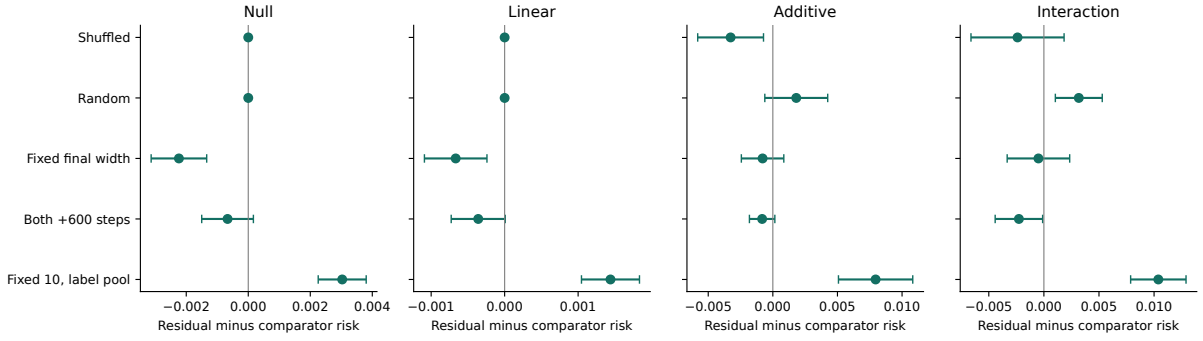


Figure 3: Selected paired comparisons from Table 2. Bars are 95% intervals across seeds. The full-label-pool comparator changes data allocation and optimizer schedule. The two tail curves continue training without the gate and have no risk-monotonicity guarantee.

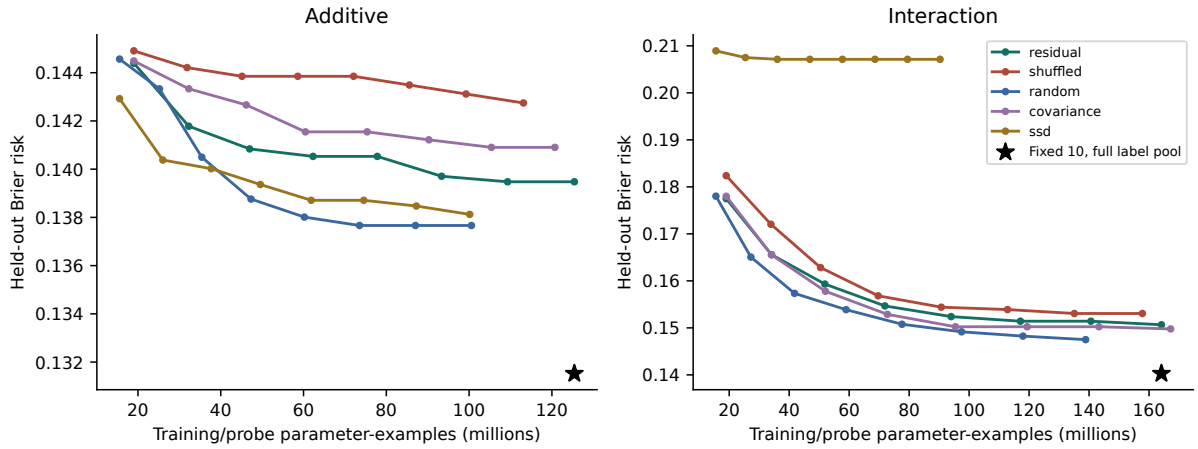


Figure 4: Measured risk versus cumulative training and candidate-probe parameter-example updates at the larger audit budget. Points summarize 16 trajectories. The fixed width-ten star uses the full label pool at the residual controller’s terminal update proxy. The horizontal axis omits audit prediction work, matrix arithmetic, optimizer constants, and memory; these are separately recorded and this plot is not a matched-FLOP comparison.

Rejection is partly a property of the chosen certificate. For the primary residual method, 253 small-audit proposals improve on both incumbent and continuation by more than τ according to the independent input evaluator; 237 are declined. At the larger audit size, 168 proposals meet this evaluator criterion and 90 are declined. These counts are diagnostic estimates, not exact population facts. The final histories differ, so the two sets are not the same fixed proposal pool.

The smaller-audit additive task contains 125 evaluator-positive proposals, all declined. Of these, 124 also have positive empirical cost-adjusted gains against both comparators; 116 would fail even a bound consisting only of the empirical-Bernstein linear term. The theoretical linear floor is $14 \log(960)/(3(n_A - 1))$, approximately 0.007825 at $n_A = 4096$ and 0.001956 at $n_A = 16384$, before accounting for variance and the growth margin. Consequently these failures cannot be presented as empirical confirmation of a minimax information lower bound. They demonstrate conservatism of this valid sufficient bound in the realized training regime.

Prediction movement and evaluator gain are separately logged for every pair. On positive nonlinear proposals their ratios are often moderately above the Bayes-refit identity $S = \Delta$, rather than uniformly exhibiting extreme cancellation. Figure 5 displays the full spread including rejected positive proposals. A small gain alone does not diagnose whether the proposal is weak, the empirical evidence is noisy, or the implemented bound is unnecessarily loose.

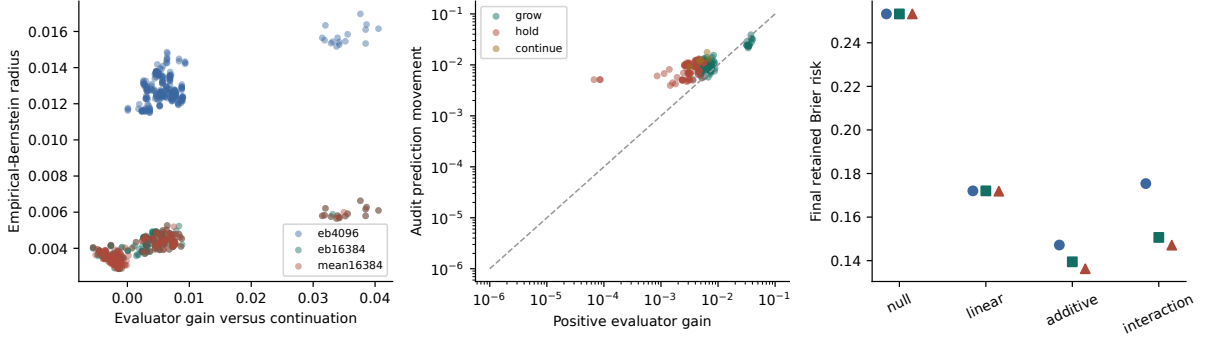


Figure 5: Measured evidence diagnostics. Left: the finite-bound radius versus evaluator gain over continuation for nonlinear residual proposals; the mean-gate radius is diagnostic only. Middle: audit prediction movement versus positive evaluator gain at the larger finite-bound audit; the dashed line is the Bayes-refit identity, not a fitted neural law. Right: final risk under small-audit finite bounds (blue), large-audit finite bounds (green), and the large-audit empirical-mean heuristic (red). All input-population quantities are Monte Carlo estimates; decisions use sampled audit labels.

The heuristic mean gate accepts 97 residual growth events and reaches mean widths 4.0625 and 6 on additive and interaction data, with risks 0.13629 and 0.14713. It provides an attainable but uncertified reference. Twenty of its retained steps have a positive evaluator risk change, all smaller than 0.001. No residual finite-bound retained step has an evaluator risk increase. Neither observation replaces the population guarantee or constitutes an exact test of its coverage; the evaluator still samples inputs.

The supplied-module evidence experiment. Separate from hidden-parameter training, 16,000 exact-distribution Monte Carlo repetitions implement the tests in Theorems 4.1 and 4.2, with $K \in \{2, 4, 8, 16\}$, $\varepsilon = 0.2$, and family error target 0.05. Signal, mixed, no-signal, and risk-neutral boundary laws are retained. Negative-binomial waiting counts, multinomial input counts, and binomial label counts are sufficient-statistic simulations of ordinary iid sampling; they do not supply exact label expectations to the tests. Module knots and amplitudes are nevertheless given, as stipulated by the theorem, and no hidden parameters are trained.

At $K = 16$ on the signal law, the reset procedure uses a mean 295,409 labels versus the shared procedure’s fixed 41,354, a factor 7.14 with these conservative constants. Family correctness is 0.998 versus 1.000 over 1,000 repetitions; zero observed errors do not imply zero error probability. At $K = 2$ the shared sufficient bound costs more, 3,506 versus mean 2,952, which is retained in the plot. Direct ReLU, Brier-risk, variance, and KL identity checks agree to $5.6 \cdot 10^{-17}$. These validate the finite realization and simulation; the lower-bound proof supplies the scaling claim.

7 Constructive follow-up: improve the gate before blaming the data

7.1 A valid alternative to the conservative EB gate

The range-dependent term in the empirical Bernstein radius is a sufficient-bound cost, which need not be unavoidable. We therefore also test a standard bounded-mean betting construction [10, Section 4]. This changes the evidence mechanism while keeping proposals and training fixed; it is not a new information score or a new statistical test.

For one comparison with required gain $\tau \in [0, 1]$, put $Z_i = D_i - \tau$, where $D_i \in [-1, 1]$ is the paired Brier difference. Our implemented thresholds are $\tau = 0$ for continuation and $\tau = 0.0005$

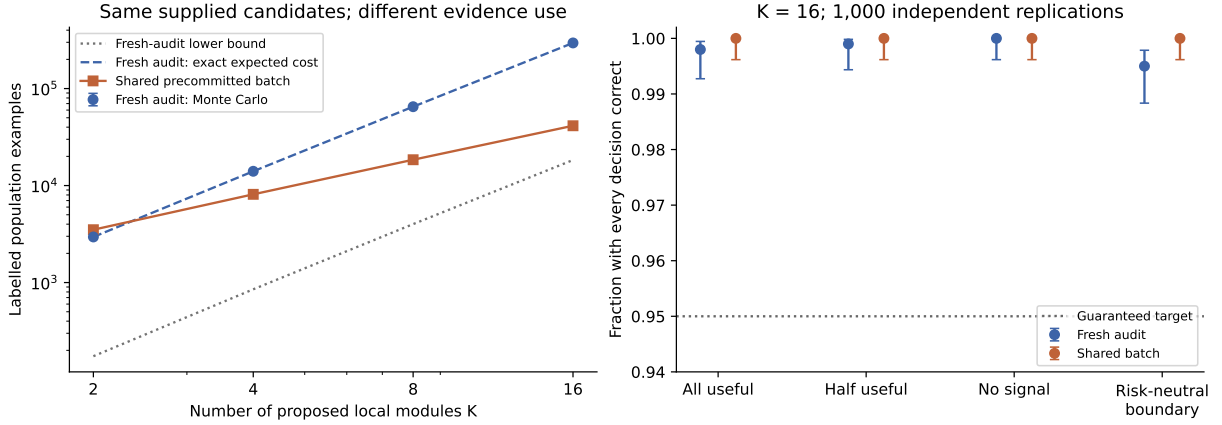


Figure 6: Supplied-module audit experiment, not ordinary neural training. Theoretical expected costs and lower bounds are distinguished from 1,000-run Monte Carlo means. Error intervals retain uncertainty at zero observed failures. The shared batch is valid because every compared local action is precommitted, whereas the reset interface withholds off-action observations.

for growth. Fix

$$\Lambda_\tau = \{0.05, 0.1, 0.2, 0.5, (1 + \tau)^{-1}\}, \quad E_n = \frac{1}{5} \sum_{\lambda \in \Lambda_\tau} \prod_{i=1}^n (1 + \lambda Z_i).$$

For these thresholds every $\lambda \in \Lambda_\tau$ belongs to $[0, (1 + \tau)^{-1}]$. The gate accepts when $E_n \geq 1/\alpha_{\text{pair}}$. It computes each product in logarithms and combines them by log-sum-exp; zero factors give a zero product. The average over the five fixed values is essential: selecting the best product without its mixture penalty would not have this validity statement.

Proposition 7.1 (Conditional validity of the implemented betting gate). *For the implemented thresholds $\tau \in \{0, 0.0005\}$, conditional on an arbitrary training and decision history H , suppose the compared predictions and τ are fixed, the audit observations are iid, and $\mathbb{E}[D_i | H] \leq \tau$. Then $\mathbb{P}(E_n \geq 1/\alpha_{\text{pair}} | H) \leq \alpha_{\text{pair}}$. The same bound holds for crossing the threshold at any audit sample size, provided the prediction pair stays fixed. Allocation $\alpha_{\text{pair}} = 0.05/(3 \cdot 8)$ gives probability at most 0.05 of any false accepted inequality during the eight-opportunity, three-comparison implemented controller, with fresh audit data at each opportunity.*

Proof. Write $M_{\lambda,0} = 1$ and $M_{\lambda,k} = \prod_{i=1}^k (1 + \lambda Z_i)$. Since $Z_i \geq -1 - \tau$ and $0 \leq \lambda \leq (1 + \tau)^{-1}$, every factor is nonnegative. Conditional on H and the first $k - 1$ audit observations,

$$\mathbb{E}[M_{\lambda,k} | H, Z_1, \dots, Z_{k-1}] = M_{\lambda,k-1}(1 + \lambda \mathbb{E}[Z_k | H]) \leq M_{\lambda,k-1}.$$

Thus each product and their fixed average form nonnegative supermartingales, with initial value one. In particular $\mathbb{E}[E_n | H] \leq 1$, and Markov's inequality proves the fixed-size claim. For the threshold-crossing claim, let T be the first crossing. For every finite m , the stopped average $E_{T \wedge m}$ has expectation at most one: this follows by summing its conditional nonpositive increments up to the bounded stopping time. On $\{T \leq m\}$ it is at least $1/\alpha_{\text{pair}}$, so $\mathbb{P}(T \leq m | H) \leq \alpha_{\text{pair}}$. Let $m \rightarrow \infty$. Finally integrate the conditional comparison bounds over H and sum them over the 24 possible tests. The three comparisons may share the current audit sample; their independence is not needed. $\square$

The proposition guarantees accepted inequalities, not the discovery of useful candidates or high power. Products are restarted for each new prediction pair. Reusing a product after a feedback-dependent change in the prediction pair is outside this implementation and proof. The

Task	Proposal	EB risk	Betting risk	Paired difference [95% CI]	Width (EB / bet)
null	residual	0.25312	0.25312	+0.00000 [+0.00000, +0.00000]	2.00 / 2.00
null	random	0.25312	0.25312	+0.00000 [+0.00000, +0.00000]	2.00 / 2.00
linear	residual	0.17210	0.17210	+0.00000 [+0.00000, +0.00000]	2.00 / 2.00
linear	random	0.17210	0.17210	+0.00000 [+0.00000, +0.00000]	2.00 / 2.00
additive	residual	0.14698	0.13836	-0.00861 [-0.01078, -0.00644]	2.00 / 3.44
additive	random	0.14698	0.13915	-0.00782 [-0.01068, -0.00496]	2.00 / 3.25
interaction	residual	0.17558	0.15209	-0.02349 [-0.02628, -0.02070]	3.00 / 5.31
interaction	random	0.17357	0.14955	-0.02403 [-0.02629, -0.02177]	3.00 / 5.56

Table 3: New-seed gate comparison, 16 paired repetitions per task and proposal. Both gates use 4,096 fresh audit labels per opportunity. Paired difference is betting minus EB Brier risk, with a descriptive 95% Student- t interval. Width is the mean final hidden-unit count. The betting gate’s extra products are not included in training parameter–example matching.

gate adds $15n$ factor evaluations across the three comparisons per opportunity, in addition to the branch fitting and prediction evaluations shared with the EB procedure. The follow-up reports these extra operations.

7.2 Fresh-seed comparison at the smaller audit budget

After the original confirmation exposed the empirical-Bernstein linear floor, we froze a separate protocol before running seeds 63000–63015. Both the betting and original EB gates use the same ordinary starts, fitting data, fresh 4,096-label audit blocks, residual/random candidate proposals, and branch training. This adds 128 task–seed–gate configurations, 256 controller trajectories, and 2,048 opportunities. It is a new-seed follow-up motivated by an earlier finding, not a retroactive change to the original confirmation. The first experiment runner and all original results remain unchanged.

The alternative gate provides a substantial constructive improvement for both proposal mechanisms. For the residual method, additive risk decreases from 0.14698 under EB to 0.13836 under betting, a paired change of -0.00861 with interval $[-0.01078, -0.00644]$. Interaction risk decreases from 0.17558 to 0.15209, a paired change of -0.02349 with interval $[-0.02628, -0.02070]$. Mean widths increase from 2 to 3.4375 on additive data and from 3 to 5.3125 on interaction data. Null and linear networks remain at width two with identical predictions under both gates. These are repeated trained histories with a finite-sample commitment guarantee, rather than forced-growth examples.

The improvement is at the same audit-label budget, not the same final training effort. Accepted wider models require more later branch training: residual training/probe parameter–example work rises by factors 1.206 and 1.224 on additive and interaction data. The betting calculations add $15 \cdot 4096 \cdot 8 = 491,520$ factor evaluations per trajectory, including rejected comparisons. Current research instrumentation also computes EB statistics in both arms. Both consume 32,768 audit labels plus 2,048 fitting labels; no evidence-sharing gain from Theorem 4.2 is claimed for this restarted testing procedure.

The follow-up also sharpens the adverse selector evidence. Under the betting gate, residual minus random is -0.000789 with interval $[-0.003284, 0.001705]$ on additive data and $+0.002542$ with interval $[0.000242, 0.004841]$ on interaction data. Random thus retains a favorable interaction result while avoiding candidate optimization. Comparisons of the residual method with its retrospectively matched fixed capacity and training/probe budget remain unresolved on both nonlinear tasks. The random betting method has a modest favorable additive comparison against its own matched fixed model, -0.00194 with interval $[-0.00353, -0.00035]$; that positive control result is retained. Larger full-label-pool networks still have lower mean risks, with the capacity and offline-optimization caveats described earlier.

Independent checks reconstruct all 1,024 betting-arm opportunities and 3,072 product tests

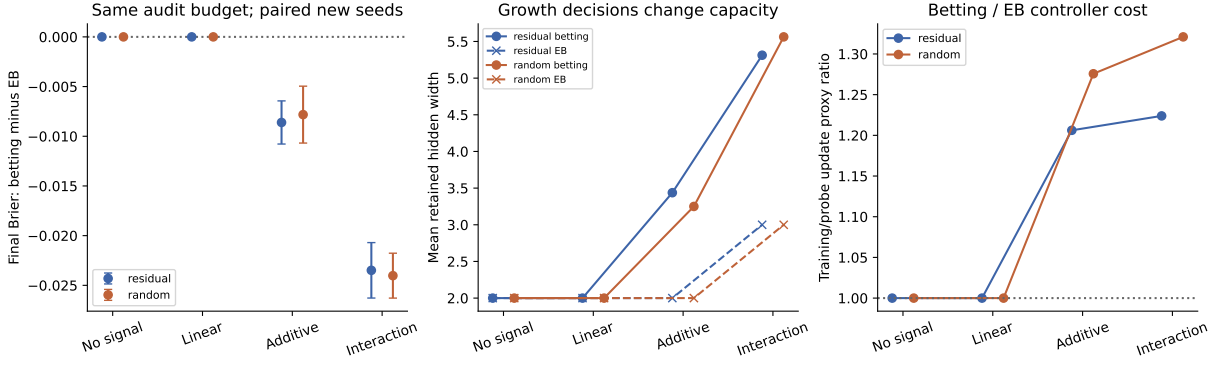


Figure 7: Measured, fresh-seed gate follow-up at the same smaller audit budget. Paired intervals and trajectories show that a less conservative established test can enable repeated growth. Extra accepted capacity raises subsequent training work. Both informative and random proposals benefit, so the result attributes progress to the evidence mechanism rather than validating the residual proposal score.

from saved parameters and regenerated audit examples; maximum log-e discrepancy is $4.0 \cdot 10^{-13}$. Finite-binomial null calculations check the expectation and tail bound directly. No betting-arm retained step increases evaluator risk in this follow-up, but that finite observation is not the proof of validity. The proof is Proposition 7.1, and evaluation remains Monte Carlo in the input distribution.

8 Interpretation and unresolved scientific questions

The implemented process can repeatedly grow and then refrain from growth in one continuing neural training history. It uses ordinary initialization, sampled labels, and trainable hidden units. This is a concrete step beyond Phase II’s one-event evidence. Yet the experiment does not establish a generally superior growth principle, nor does it learn its opportunity schedule, action family, or risk threshold. The useful empirical distinction is between a score’s ability to generate good proposals and a gate’s ability to identify their benefits at an affordable resource cost.

The theory makes the same distinction more sharply. Brier movement and gain bound paired-loss variance and characterize worst-case testing effort. The module theorem shows that discarding off-action evidence can create a quadratic serial label cost even when all supplied operations are useful. Shared precommitted tests and targeted labelling defeat that cost in the same population. Thus an important design variable is the evidence the learner retains, not only the statistic computed for an individual unit. The lower bound is not an unavoidable tax on the actual supervised network: its fitting history contains information deliberately absent from the theorem’s restricted interface.

The implemented finite-sample gate may also be conservative. Its range-dependent linear radius does not vanish as rapidly as actual movement in every comparison. Observed rejection of evaluator-positive candidates therefore does not prove that every valid test would need the same number of labels. The theoretical lower bound and the gate’s numerical constants are different statements. More efficient valid tests, retained-data procedures, and jointly evaluated action batches could improve this controller without contradicting any result here.

No-growth has a further boundary independent of statistics. A useful joint change can be assembled from singleton actions whose cost-adjusted gains are too small to accept separately. The exact calculation and finite-sample diagnostic in Appendix A make this explicit with full observations and convex decoder fitting. This is established conditioning/complementarity mathematics [3, 4], not a new universal impossibility theorem. It shows why avoiding all individually questionable steps is not the same as finding a good structure.

The important open version of the original question remains a learner that discovers useful nonlinear operations, shares past evidence legitimately, and improves the risk–capacity–data–computation tradeoff across ordinary tasks. There is no proof here of learned semantic units, efficient online mutual information estimation, optimal stopping over all architectures, automatic discovery of the theorem’s modules, or benefits on real-world deep networks. The constructive progress and the negative attribution evidence should both inform that next investigation.

9 Reproducibility and claim status

The complete project is under `phase3/`. The earlier phases remain unchanged and are identified by SHA-256 manifests. Both exploratory pilots, all new-seed trajectories, trained branch parameters, proposal pools, gate statistics, event decisions, resource proxies, and configurations are saved. Random streams reconstruct all fitting, audit, and evaluation examples. The main runner is `code/sequential_growth.py`; scripts regenerate all tables and figures from those saved outputs. Full command lines and source hashes accompany each main result. Separate audit scripts check gradients, function preservation, conditional data flow, gate replay, and saved-result consistency. They are implementation evidence, not a substitute for the complete proofs above.

The inherited findings, known lemmas, project-derived theorems, conjectural interpretations, and empirical outcomes are mapped in `EVIDENCE_LEDGER.md`. Primary-source notes record inspected theorem and algorithm locations and versions. A Chinese explanation and `STATUS.md` summarize the actual results and the next concrete research step. Local NumPy/SciPy CPU computation and existing LaTeX/Poppler tools suffice; no paid compute or external private-data service is used.

A Why a rejected singleton need not mean sufficient structure

This auxiliary investigation was developed independently of the main controller. It isolates an action-horizon boundary with full observations and exact convex refitting. The mechanism is a specialization of known weak-submodularity and covariance-conditioning results [3, 4]; it is not claimed as a new general impossibility theorem.

Let U, V, E be independent standard Gaussians, $Y = V + \sigma E$, and let the available frozen features be $h_+ = U + \epsilon V$ and $h_- = U - \epsilon V$, where $0 < \epsilon < 1$. A structural action adds features and exactly refits an affine least-squares decoder. Let $F(S)$ be its population mean-squared risk reduction relative to the intercept-only optimum.

Proposition A.1 (Complementary actions under a per-feature charge). *One has*

$$F(\{+\}) = F(\{-\}) = \frac{\epsilon^2}{1 + \epsilon^2}, \quad F(\{+, -\}) = 1.$$

For $\epsilon^2/(1 + \epsilon^2) < \lambda < 1/2$, every rule that requires each singleton addition to decrease risk plus λ times width stays at width zero, however many times the same opportunities recur. A two-feature addition decreases that objective by $1 - 2\lambda > 0$.

Proof. The feature variances equal $1 + \epsilon^2$, while their covariances with Y are ϵ and $-\epsilon$. Minimizing the quadratic $\mathbb{E}(Y - bh)^2$ gives $b = \mathbb{E}[Yh]/\mathbb{E}[h^2]$ and reduction $(\mathbb{E}[Yh])^2/\mathbb{E}[h^2]$. This proves the singleton formulas. The pair can represent $V = (h_+ - h_-)/(2\epsilon)$, leaving only independent noise with risk σ^2 . The initial risk is $1 + \sigma^2$, so the joint reduction is one. Subtracting the specified feature charges proves the claims. $\square$

The singleton gains are positive. This is a cost-threshold obstruction, not a claim that a population gate with zero threshold must reject them. The standardized feature correlation is $(1 - \epsilon^2)/(1 + \epsilon^2)$, so its smallest covariance eigenvalue is $2\epsilon^2/(1 + \epsilon^2)$, exactly the sum of singleton

gains divided by their joint gain. This is the established submodularity-ratio mechanism. If a candidate dictionary has ratio at least $\gamma > 0$, all singleton gains at most τ imply a block of at most k features has gain at most $k\tau/\gamma$: sum the singleton bounds and apply the ratio definition. Without such conditioning the conclusion degenerates.

The pair decoder pays a further resource: its outgoing coefficients are $(1/(2\epsilon), -1/(2\epsilon))$, with ℓ_1 norm $1/\epsilon$ and squared ℓ_2 norm $1/(2\epsilon^2)$. Adding a penalty $\rho\|b\|_2^2$ changes the optimal penalized gain to $2\epsilon^2/(2\epsilon^2 + \rho)$. Indeed, antisymmetric coefficients $(a, -a)$ minimize $(1 - 2\epsilon a)^2 + 2\rho a^2$ at $a = \epsilon/(2\epsilon^2 + \rho)$. Substitution proves the formula. Thus large cancellation and poor conditioning are not free.

In a separate binary-label prediction problem, an information decomposition also cautions against promoting a representation score into realized learning benefit. Under logarithmic loss, for current observations H , candidate observation Z , and fitted predictions $q(H), q'(H, Z)$,

$$R(q) - R(q') = I(Y; Z \mid H) + A - A',$$

where $A = \mathbb{E}D_{\text{KL}}(P(Y \mid H) \parallel q(H))$ and $A' = \mathbb{E}D_{\text{KL}}(P(Y \mid H, Z) \parallel q'(H, Z))$. This is the standard conditional cross-entropy identity: condition each risk on its available observations, obtain conditional entropy plus KL approximation error, and subtract. CMI equals realized improvement only when those decoder errors cancel, in particular for two exact Bayes refits. The neural experiment uses Brier loss; this identity is an interpretative result, not its score.

A.1 Executed finite-sample action-horizon diagnostic

The separate diagnostic offers twelve fixed features: four independent complementary pairs and four independent nuisances. Signal targets combine four independent V coordinates with coefficients $(1, 0.8, 0.6, 0.4)$ plus Gaussian noise of standard deviation 0.5; a null condition uses noise only. For each of 32 paired seeds, $\epsilon \in \{0.1, 0.25\}$, and singleton or up-to-pair actions, 1,024 labels fit the readout. Ten opportunities each receive 512 fresh validation labels. All singleton or singleton-and-pair candidates are fitted, then the best positive empirical validation gain minus 0.03 per added feature is selected. The controller receives no latent pairing identifiers or population risk. This gate is heuristic: it maximizes uncorrected validation scores and Gaussian squared losses are unbounded. Proposition 5.1 does not apply.

There are 256 executed configurations. At $\epsilon = 0.1$ on the signal task, mean exact-population MSE is 1.6916 for singleton actions and 0.2520 for pair actions; final widths average 1.81 and 8. At $\epsilon = 0.25$, mean MSEs are 0.4405 and 0.2520. Bayes risk is 0.25. Both policies remain at width zero in all null configurations. Every fitted readout and candidate score is saved. These deliberately complementary frozen features demonstrate an available-action boundary, not general neural proposal learnability.

The pair policy pays substantially more. At $\epsilon = 0.1$ signal, mean readout fits are 251 versus 111.1, and the $n(k+1)^2$ least-squares proxy is 12.23 million versus 1.06 million. On null data the pair policy fits 781 readouts versus 121, including the initial intercept fit, while rejecting all growth. Coefficient norms are recorded. An exact-population singleton cost gate would reject all first moves at $\epsilon = 0.1$, yet its empirical counterpart grows in 27 of 32 signal runs: noise sometimes allows a weak first step whose complement later helps. This is not a recommendation to accept false positives. It shows why a strict per-step objective and an inadequate action horizon may conflict.

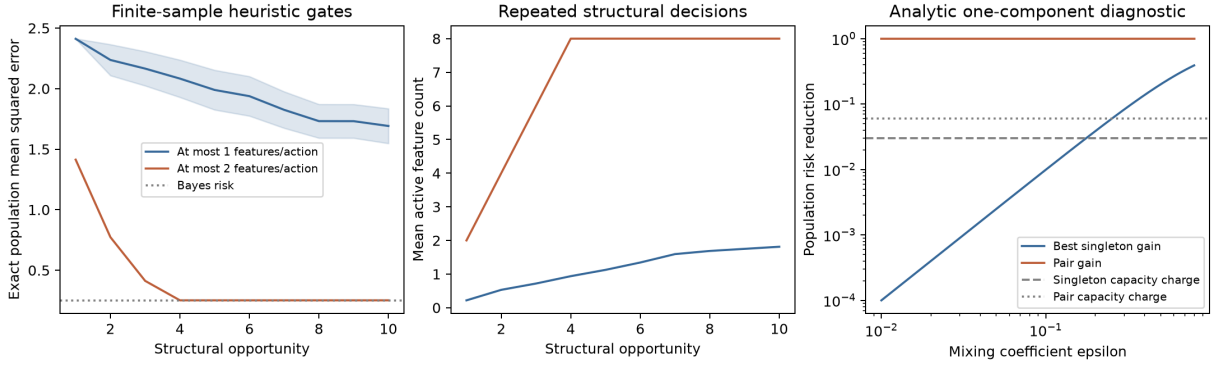


Figure 8: Auxiliary frozen-feature/readout experiment, separate from neural hidden-parameter training. Analytic one-component quantities and measured repeated trajectories are labelled separately. Bands are pointwise 95% Student- t intervals over 32 independent runs; the exact population evaluator does not enter selection. Pair enumeration uses more candidate fits and may require large cancelling output coefficients.

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